

Chitwan Saharia

AI Resident, Google Brain, Toronto

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Education

B.Tech. with Honors in Computer Science and Engineering

Indian Institute of Technology, Bombay, Mumbai, India

2015-19

GPA 9.11/10

Interests

Machine Learning, Natural Language Understanding, Reinforcement Learning

Pre-Prints

- **Non-Autoregressive Machine Translation with Latent Alignments**
*Chitwan Saharia**, William Chan*, Saurabh Saxena, Mohammad Norouzi
- **Imputer: Sequence Modelling via Imputation and Dynamic Programming**
William Chan, *Chitwan Saharia*, Geoffrey Hinton, Mohammad Norouzi, Navdeep Jaitly

Publications

- **Combating False Negatives in Adversarial Imitation Learning for Instruction Following**
K. Zolna*, *Chitwan Saharia**, L Boussieux*, David H, Maxime CB, D Bahdanau, Yoshua Bengio
AAAI Conference (Student Abstract Track), 2020
- **A Tale of Two Modalities for Video Captioning**
P. Joshi*, *Chitwan Saharia**, Vishwajeet Singh, D. Gautam, G. Ramakrishnan, P. Jyothi
Workshop on Multi-modal Video Analysis & Moments in Time Challenge, ICCV, 2019
- **BabyAI: First Steps Towards Grounded Language Learning With a Human In the Loop**
Maxime CB, Dzmitry Bahdanau, Salem L, Lucas W, *Chitwan Saharia*, Thien HN, Yoshua Bengio
International Conference on Learning Representations (ICLR) 2019
* Equal Contribution

Internships and Projects

- **Grounded Language Learning with a human in the loop** **MILA, Montreal**
Internship, Advisors : Dr. Yoshua Bengio and Dzmitry Bahdanau *May 2018 - August 2018*
 - Worked on implementing **BabyAI open source platform** for training instruction following artificial agents on levels using heuristic expert agent for the purpose of simulating a human teacher.
 - Reported baseline results and estimate of the amount of human involvement that would be required to train a neural network-based agent on some of the BabyAI levels.
 - Implemented variety of algorithms including vanilla imitation learning, active imitation learning and behavioural cloning supported by automated curriculum to train multiple levels simultaneously.

- **Learning Shared Representations from Audio and Visual Data [Talk]** **IIT Bombay**
Bachelor's Thesis, Advisors : Prof. P. Jyothi and Prof. Ramakrishnan *August 2018 - Present*
 - An ongoing IBM AI Horizons project on learning rich embeddings from video data using cross modal attention on audio and video stream for video captioning.
 - Analysing effect of cross modality fusion techniques, pre-training of encoders, fine tuning of model using sequence based RL loss, and importance scaling across modalities on captioning performance from qualitative point of view.
- **Calibration of Neural Networks for Cell Segmentation [Report]** **IIT Bombay**
Research Project, Advisor : Prof. Suyash Awate *August 2018 - Present*
 - Implementing different calibration techniques like Platt Scaling, Entropy Regularization, Binning, Temperature Scaling and examining their performance calibrating cell segmentation tasks.
 - Examining the effect of calibration on test performance of noisy cell segmentation datasets.
 - Exploring use of assumed density filtering to propagate uncertainty through network activations.
- **Hierarchical Dialog Models for Amazon Alexa Challenge** **MILA, Montreal**
Research Internship, Advisor : Dr. Yoshua Bengio *May 2017 - August 2017*
 - Implemented dialog generative model as a part of Amazon Alexa Challenge using hierarchical self-attention encoders to propagate long term contexts in dialog agents.
 - Explored use of tf-idf retrieval model as an auxiliary input along with attention weighted encoder states to improve performance over baseline model.
- **Sentiment Analysis on Social Media [Report] [Talk]** **IIT Bombay**
Seminar, Advisor : Prof. Pushpak Bhattacharyya *January 2018 - April 2018*
 - Explored traditional machine learning methods for sentiment analysis and compared their performance with rule based lexical methods on social media setting.
 - Explored leveraging of social media specific heuristics like punctuations, capitalization, intensifiers, sentiment shifters for improving rule based sentiment analysis models.

Scholastic Achievements

- Secured All India Rank 148 in IIT JEE Examination (2015) among 150 thousand candidates.
- Secured 99.9 percentile in JEE-Main (2015) out of 1.5 million candidates.
- Recipient of Kishore Vaigyanik Protsahan Yojna Fellowship instituted by the Department of Science and Technology, Govt. of India.

Key Academic Projects

- **Deep Kalman Filters for Time Series Prediction**
 Implemented Deep Kalman Filters as a part of Kaggle Challenge for time series prediction of Rossman Sales Data. Implemented base line models including vanilla LSTMs, **scheduled sampling** and compared its performance with them.
- **Analysis and Improvement of Speaker Identification Models**
 Worked on analysis of state of the art text independent open set speaker recognition models using CNNs and comparing their performance with the **i-vector** models. Used **Voxceleb** dataset for training and evaluation.
- **Analysis of Image Captioning models**
 Implemented state of the art image captioning papers - **Show & Tell**, and **Show, Attend & Tell**, and compared their performance. Used pretrained VGG-19 network as an encoder and

fine tuned it together with the LSTM decoder.

- **Deep Q Learning Agent for Flappy Bird**

Trained a flappy bird game bot using Deep-Q Learning and compared its performance with state of the art supervised learning models. Used Support Vector Machines and Neural networks as baseline model. Analysed the performance of neural network models using different hidden units, layers, activation functions and gradient optimizers.

- **Intelligent Pacman Agent**

Developed an agent to play Pacman utilizing heuristic based Minimax and Expectimax search with Alpha-Beta pruning for tackling multiple ghosts. Used Particle Filters and Dynamic Bayesian Nets for tracking the approximate positions of the ghosts.

- **Other Projects**

- **Compiler for Pointer Language** : Designed a python based compiler for a C like pointer language with support for scopes, procedures, iteration, conditionals and function calls.
- **ATM Controller** : Developed an ethernet connected ATM controller network using VHDL with C supported backends supporting caching, and offline syncing with the server.
- **Chatting Application** : Developed a central server based chatting application in C++ with graphics supported by ncurses library supporting text and file transfer features.

Positions of Responsibility

- **Teaching Assistant**

- *Object Oriented Programming at IIT BombayX* : Responsible for preparing walk throughs and tutorials for the course, and responding to discussion forum queries for around 1000 students.
- *Computer Programming at IIT Bombay* : Responsible for preparing lab questions, and managing graded lab sessions for 400 1st year students.

Relevant Academic Coursework

- **Machine Learning** - Foundations of Machine Learning, Automated Speech Recognition, Probabilistic Graphical Models, Digital Image Processing, Foundations of Intelligent Learning Agents, Artificial Intelligence, Data Analysis and Interpretation.
- **Mathematics** - Calculus, Linear Algebra, Differential Equations, Numerical Analysis.
- **Computer Science** : Discrete Structures, Data Structure and Algorithms, Database Systems, Operating Systems, Implementation of Programming Languages, Automata Theory.